LOCOMOTIVE BOM (parts grouped by CAD name; masses = model volume × 7,850 kg/m³, steel only)

Item	Description (from CAD model PF-L100)	Qty	Material	kg (model)
1	main rail HSS12x6x3/8 (LT18: Y±240, Z1150-1454.8)	2	A500 Gr C	1,389
2	front end sill HSS12x6x3/8	1	A500 Gr C	21
3	rear end sill HSS12x6x3/8	1	A500 Gr C	21
4	crossmember HSS8x4x1/4	5	A500 Gr C	48
5	body bolster box PL12/PL10	2	A36	78
6	body centre plate Ø300 PL40 (truck top Z1110, LT18)	2	A36	44
7	running-board outrigger C3x4.1	14	A36	554
8	running-board edge angle L3x3x1/4	2	A36	230
9	running board / deck PL6 tread plate (top Z1,605, LT11)	1	A36	1,248
10	loco rear drop drawbar box PL16	1	A572 Gr 50	155
11	loco rear pocket top plate PL20	1	A572 Gr 50	14
12	loco rear pocket bottom plate PL20	1	A572 Gr 50	14
13	pilot / buffer beam PL12 box	1	A36	-
14	ballast pocket PL12 + shear lugs (750 kg max each side)	2	A572 Gr 50	128
15	ballast plates 800x480x25 A36, up to 9 per side (removable)	2	A36	1,441
16	axle Ø1450 driver	4	EA4T / 4140	467
17	wheel tread rim 50 deep Ø1450 driver	8	EN 13262 ER7 / AAR B	1,657
18	wheel web 25 Ø1450 driver	8	EN 13262 ER7 / AAR B	2,198
19	wheel hub OD200 Ø1450 driver	8	EN 13262 ER7 / AAR B	137
20	flange Ø1450 driver	8	EN 13262 ER7	182
21	shrunk sprocket hub 4140 Ø240x110 + Ø350 flange on raised Ø150 axle seat, SHRI…	4	AISI 4140 Q&T	95
22	hub flange Ø350 x 20 integral (L4.1), 12 x Ø17 reamed on PCD 300	4	EN 13262 ER7	32
23	45T ANSI 120-2 split rim sprocket ring with web PD546 OD569 (Source Bot rev 3)…	4	purchased	-
24	oil-bath chain case 3 mm + floating seal plate at the axle (ISO VG 220, ~4 L)	4	A36 3 mm + seal plate (L-103.2)	-
25	15T ANSI 120-2 motor sprocket Martin D120B15 PD183 OD202, hub broached DIN 548…	4	purchased	-
26	Poclain MS05 468/234 cc 2-speed HighFlow + T04 brake, shaft code 5 DIN 5480 W5…	4	purchased	-
27	motor bracket PL25 slotted (4 x M20 in 50 mm slots, 2 jack screws) welded to t…	4	A572 Gr 50 PL25 (L-103.2)	127
28	motor bracket gusset/top plate PL20 to transom	4	A572 Gr 50 PL20 (L-103.2)	5
29	brake disc Ø340x25 on the axle (Y+, unchanged)	4	purchased	-
30	spring-applied air-released caliper (2 per disc, ref. Coremo D 3.5N)	8	purchased	-
31	inside axlebox + primary guide (AWAIT bearing)	8	purchased	-
32	primary spring Ø140 reservation (LT35 AWAIT)	8	purchased	-
33	crank pin boss (decorative)	8	A36 (decorative)	2
34	inside sideframe top chord PL30 (top Z1110, LT18)	4	A572 Gr 50	228
35	inside sideframe bottom chord PL30	4	A572 Gr 50	166
36	horn cheek PL30	16	A572 Gr 50	48
37	side rod (bronze look, decorative, static phase)	4	A36 (decorative)	107
38	truck transom box PL16	2	A36 / A500 Gr C	155
39	truck bolster / pivot seat PL16	2	A36	17
40	traction pin Ø120 4140 Q&T, Z650-1110 (L4.2)	2	AISI 4140 Q&T	82
41	traction-pin bearing housing GE120ES (AWAIT SKF rating) at Z650 under transom …	2	purchased	-
42	front-truck cylinder-support extension PL30	2	A572 Gr 50	45
43	cylinder support arm PL25 (lengthened for Y±790)	2	A36	30
44	decorative cylinder Ø380 at Y±790 (moved from ±750:-wheel clash K2-h, route ch…	2	A1011 CS (decorative)	-
45	axle	2	EA4T / 4140	191
46	wheel tread rim 50 deep	4	EN 13262 ER7 / AAR B	355
47	wheel web 25	4	EN 13262 ER7 / AAR B	162
48	wheel hub OD200	4	EN 13262 ER7 / AAR B	83
49	flange	4	EN 13262 ER7	42
50	leading axlebox (inside)	4	purchased	-
51	leading-truck inside sideframe PL25	2	A36	134
52	leading-truck transom	1	A36 / A500 Gr C	73
53	pony ballast 0.625 t per side (steel plates on the sprung pony frame, own seco…	2	purchased	-
54	leading-truck swing arm 1,075 (LT18)	1	A36	49
55	swing-arm anchor hanger PL20 box	1	A36	-
56	boiler-style hood shell 3 mm (cosmetic, removable)	1	A1011 CS 3 mm	-
57	smokebox front	1	A1011 CS	-
58	stack (cosmetic)	1	A1011 CS	-
59	dome (cosmetic)	1	A1011 CS	-
60	cab lower walls 3 mm on HSS2x2x1/8	1	A500 Gr C	-
61	cab front post HSS4x4x1/4	2	A500 Gr C	215
62	cab rear post HSS4x4x1/4	2	A500 Gr C	215
63	cab front wall / windows	1	PC glazing on A500 frame	-
64	cab roof 3 mm skin on HSS3x2x1/8 purlins @600 (3 in deep, adopted R2)	1	A1011 CS skin on HSS3x2x1/8 purlins	
65	cab step tread PL8 grip strut	8	A36	60
66	step stringer FB 20x75	8	A36	113
67	grab iron Ø32 solid A36	4	A36	30

Purchased items: mass per supplier (AWAIT). '-' = shell / wall / roof / envelope modelled solid in CAD - mass from the ledger, not the model. L4 drive units (Poclain MS05 + T04, RS120-2 chain 15T:45T, pressed hub, oil-bath case, disc/2 calipers) detailed on L-103.2 (×4; shop detail Loco Bot L-103). Front ballast 150.8 kg (2 × 75.4 kg plates, 1 per side) + BT-1 battery 68.2 kg in the Y… ell lec
CAD steel total PF-L100: see cad/mass_report.json. Loco ledger rev L4.2 incl. barrel 2,280.6, traction pins Ø120 (+22 kg each vs Ø100), NB-1 59.8 kg, final drive 336.7 kg/axle, ballast and pony ballast: 19,934 kg (drive_L4.json; L-107 supports pending).

PE STAMP / SIGNATURE

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UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Bill of materials - locomotive PF-L100 (rev L4.2)

Sheet S-501 · REV L4.2 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED